Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**A Real-to-Sim-to-Real Technology for Reliable Urban Autonomous Navigation in Long-Tail Scenarios**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (√), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (), Cybersecurity (), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ()			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	ITS / Telematics	ITS Application Services
Concept of Proposed Technology	The proposed technology aims to develop **a Real-to-Sim-to-Real (R2S2R) framework that enables systematic validation and improvement of the reliability of urban autonomous navigation systems** (e.g., autonomous driving vehicles and delivery robots) in **rare, long-tail, and safety-critical scenarios** (e.g., erratic pedestrians, aggressive cut-ins, unusual road events, and adverse weather). The system leverages advanced 3D/4D computer vision technologies and generative AI to construct large-scale simulation environments that replicate real-world conditions from real-world driving data. These environments are augmented with language-guided 4D long-tail scenarios, enabling systematic stress testing and refinement of autonomous driving policies. By shifting closed-loop testing and policy			

refinement from costly and risky real-world trials to high-fidelity simulation and controlled mixed-reality evaluation, the proposed technology can significantly improve the navigation policy reliability in long-tail scenarios, while reducing validation cost, deployment risk, and on-site engineering effort.

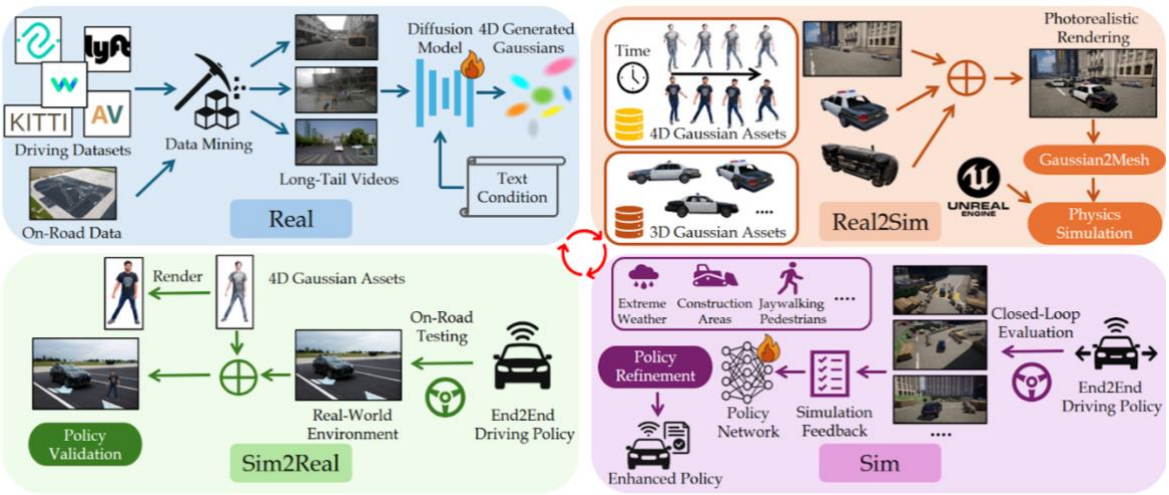

Figure 1. Overview of the proposed Real-to-Sim-to-Real framework for advancing urban autonomous navigation systems in long-tail scenarios.

As shown in the above figure, the framework has four main functions:

1. (Real) Long-tail data mining from large-scale real-world driving logs to detect, index, classify rare events, and curate a structured long-tail driving dataset.
2. (Real2Sim) 4D dynamic scenario generation to create controllable, high-fidelity long-tail scenarios from text prompts and integrate them into a photorealistic, physics-aware simulator that replicates real-world urban environments.
3. (Sim) Closed-loop validation in the high-fidelity simulator with diverse long-tail scenarios for systematic policy performance testing and refinement.
4. (Sim2Real) Mixed-reality testing that renders generated virtual dynamic assets into onboard real sensor streams in real time for real-world closed-course evaluation, bridging simulation and real-world testing safely.

The key novelty of the proposed technology is an end-to-end pipeline that transforms real-world rare events into a searchable long-tail scenario library, expands them into diverse yet controllable 4D variants, and systematically validates and refines autonomous navigation policies through both high-fidelity simulation and mixed-reality testing. The proposed technology enables **industry partners across autonomous mobility and robotics to enhance real-world reliability, safety, and deployment readiness**, while reducing validation cost and risk for autonomous driving, advanced driver-assistance systems (ADAS), and urban delivery and service robot applications.

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Rationale for Joint R&D Project	Urban autonomous navigation systems, including autonomous driving vehicles, ADAS, and urban delivery and service robots, play a critical role in next-generation mobility and intelligent transportation. However, **ensuring reliable and safe operation under rare, long-tail, and safety-critical scenarios**, such as erratic pedestrian behavior, aggressive cut-ins, unusual road events, and adverse weather, remains a **major technical barrier**. Existing development and validation pipelines rely heavily on extensive real-world testing with manual intervention, which is costly, time-consuming, and inherently risky. For industrial partners, these challenges translate into increased deployment risk, high validation costs, extended development cycles, and delayed commercialization. Meanwhile, recent advances in 3D/4D computer vision (CV), generative AI, and large-scale simulation technologies present new opportunities for **systematically addressing long-tail reliability challenges through simulation- and mixed-reality–based validation**. Nevertheless, developing such a Real-to-Sim-to-Real framework requires the tight integration of advanced CV/AI models, large-scale real-world driving data, high-fidelity simulation environments, and real autonomous platforms, along with deep domain expertise in autonomous navigation and system validation. Achieving this integration is difficult for a research institute or an industry partner to accomplish independently, making joint R&D essential for success and effective technology transfer. Rationale for Joint R&D: • The research institute contributes advanced CV and AI models for long-tail data mining, 3D/4D scene reconstruction, dynamic scenario generation, as well as simulation-based training and validation, and mixed-reality testing capabilities. • The industry partner provides real-world industrial requirements, autonomous driving or robotic platforms, operational constraints, representative driving data, and support for hardware integration and field testing. • Effective co-development, validation, and refinement of CV/AI software, simulation tools, and autonomous systems are necessary to ensure industrial relevance, reduce technical and deployment risk, and accelerate technology maturation. Expected benefits from joint R&D: • Accelerated technology transfer from advanced computer vision and generative AI research to deployable reliability validation tools for urban mobility products. • Reduced technical, validation, and deployment risk through joint development and simulation- and mixed-reality-based evaluation. • Improved industrial relevance and practicality by tightly integrating research innovations with real-world operational requirements. • Establishment of a scalable foundation for commercialization of urban autonomous robots.

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Key Words	1. Autonomous Driving in Long-Tail Scenarios 2. Urban Delivery Robots 3. Real-to-Sim-to-Real (R2S2R) 4. 4D Scene Generation 5. Mixed-Reality Testing

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop a Real-to-Sim-to-Real (R2S2R) framework that enables systematic validation and reliability improvement of urban autonomous navigation systems, such as autonomous driving or delivery robots, under rare, long-tail, and safety-critical scenarios through integrated long-tail data mining, 3D/4D scene reconstruction and generation, and simulation- and mixed-reality-based testing. The specific R&D objectives are: • To develop a Real-to-Sim-to-Real (R2S2R) framework for constructing physics- and photorealistic urban simulations from large-scale real-world driving data, enabling scalable, cost-efficient, and low-risk validation of autonomous navigation systems. • To develop a 4D generative model for controllable long-tail scenario synthesis from image and text inputs, with spatiotemporally consistent dynamic interactions integrated into simulation-based testing. • To improve the reliability of urban autonomous navigation policies in edge and long-tail cases through simulation-based training and refinement. • To develop a mixed-reality testing pipeline that enables real-time rendering of generated 4D assets into live sensor streams for closed-course evaluation prior to real-world deployment.			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Long-Tail Scenario Data Mining**: Curate ≥500 long-tail driving sequences from large-scale real-world driving data, indexed with rich metadata (e.g., scenario type, location, time, weather, and textual descriptions). The curated dataset will serve as the foundation for training and evaluating 4D generative models. **4D Scenario Generation Quality and Utility**: Generate diverse, controllable, and realistic 4D long-tail scenarios suitable for closed-loop policy testing, supporting multi-object interactions and spatiotemporal consistency. Achieve: • Temporal coherence with frame-to-frame geometric consistency error ≤ 5 cm for dynamic objects • Visual realism with PSNR ≥ 25 dB (or equivalent perceptual quality metric) for rendered views • Scenario controllability demonstrated by consistent adherence to text/image prompts across ≥ 80% of generated scenarios. **Mixed-Reality Fidelity**: Achieve <0.1 LPIPS (or equivalent perceptual metric) between real images and mixed-reality augmented outputs in controlled evaluation. **Navigation Policy Improvement**: Achieve ≥10% improvement in edge-case closed-loop metrics (including driving score, success rate, and interactive behaviors, or equivalent safety metric agreed with the industry partners) without degrading nominal performance.			

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R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy that advances from real-world data mining to simulation construction, then to simulation-based policy validation and refinement, and finally to real-world mixed-reality testing. • **Year 1: Long-Tail Data Mining and Real-to-Sim Urban Simulation Construction** **Objectives**: Establish a high-fidelity urban simulation environment from real-world driving data, develop long-tail data mining and 4D scenario generation capabilities in simulation. **Key tasks**: ○ Develop a real-to-sim framework to construct physics- and photo-realistic simulation environments that replicate the real-world urban environments from large-scale real-world driving data, leveraging 3D/4D scene reconstruction and Gaussian Splatting-based rendering. ○ Develop a long-tail data mining pipeline to detect, index, and classify rare and safety-critical events from large-scale real-world driving logs. ○ Develop and train a 4D generative scenario model from curated long-tail dataset to synthesize controllable long-tail scenarios from image and text inputs, supporting spatiotemporally coherent dynamic objects and interactions. ○ Integrate generated long-tail scenarios into the urban simulator to enable closed-loop policy testing under rare and challenging conditions. **Milestones**: ○ High-fidelity large-scale simulation environment that replicates real-world urban environments and supports closed-loop autonomous navigation policy testing. ○ Curated long-tail driving dataset and a 4D long-tail scenario generation module integrated into simulation. • **Year 2: Simulation-Based Navigation Policy Closed-Loop Testing and Training** **Objectives**: Leverage simulation and generated long-tail scenarios to systematically close-loop stress test the navigation policy reliability, and further refine the policy in simulation. **Key tasks**: ○ Conduct closed-loop testing of autonomous navigation policies across diverse long-tail scenarios in simulation, establishing quantitative performance benchmarks. ○ Develop simulation-based training and refinement strategies, including curriculum reinforcement learning and scenario-adaptive residual policies, to improve edge-case performance without degrading nominal behavior. ○ Enhance the 4D generation pipeline with scenario diversification, controllability, and domain randomization to increase robustness and coverage. ○ Optimize simulation efficiency and data pipelines to support large-scale testing and iterative policy refinement. **Milestones**: ○ Demonstrated improvement in autonomous navigation policy reliability under long-tail scenarios in simulation. ○ A mature R2S2R workflow enabling efficient policy evaluation and refinement. • **Year 3: Real-World Mixed-Reality Testing and Deployment Readiness** **Objectives**: Demonstrate deployment readiness, scalability, and industrial relevance through mixed-reality testing in real-world and prepare the technology for industrial transfer. **Key tasks**: ○ Develop a mixed-reality testing pipeline that enables real-time insertion and rendering of generated 4D dynamic assets into robot live sensor streams. ○ Conduct extensive real-world mixed-reality testing to evaluate autonomous

	navigation behavior under scenarios difficult or unsafe to reproduce physically. ○ Validate system reliability across diverse real-world urban environments and virtual long-tail scenario types. ○ Assess commercialization potential and prepare the technology for industrial transfer, including documentation, interfaces, and deployment guidelines. **Milestones**: ○ Successful demonstration of mixed-reality testing for autonomous navigation under generated virtual long-tail scenarios in real-world nominal environments. ○ Technology readiness advanced to TRL 6, with a clear pathway for SME adoption and commercialization.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the primary technical lead for the joint R&D project, focusing on the development of advanced computer vision and generative AI algorithms, Real-to-Sim-to-Real simulation technologies, mixed-reality testing capacities, and navigation policy training and testing pipelines required for improving the reliability of urban autonomous navigation systems in long-tail scenarios. Expected R&R of Research Institute: ○ Development of core computer vision and AI technologies for the Real-to-Sim-to-Real (R2S2R) framework, including long-tail data mining, 3D/4D scene reconstruction and generation, and advanced rendering methods for mixed-reality testing. ○ Development of closed-loop navigation policy testing and simulation-based training pipelines to enable systematic evaluation and improvement of policies under long-tail, safety-critical scenarios. ○ Provision of technical leadership, research expertise and resources, as well as support for system integration and technology transfer to enable deployment readiness and industrial adoption. The PI has extensive prior collaborations with autonomous driving industry partners, including Toyota Research Institute, Woven, and Ford Motor Company.
Expected R&R of Korean SME Partner	The Korean SME partner will play a critical role in ensuring industrial relevance, practical feasibility, and successful deployment of the proposed technology by contributing real-world autonomous systems, operational expertise, and validation environments for urban autonomous navigation. Specifically, the Korean SME partner will be responsible for: • Defining real-world operational requirements and representative use cases, including environmental factors, safety constraints, and deployment workflows, to guide system design and performance evaluation. • Providing autonomous navigation platforms, sensing systems, and system integration support, such as autonomous driving vehicles, ADAS-equipped vehicles, or urban delivery and service robots used in real-world operation. • Supporting real-world data collection, hardware deployment, and mixed-reality testing in closed-course or controlled environments. • Leading commercialization planning and industrial transfer, including assessment of deployment feasibility, regulatory and safety requirements, and pathways for integrating the developed R2S2R technology into existing autonomous mobility and robotics products or services.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes:** • A mature Real-to-Sim-to-Real (R2S2R) pipeline that transforms real-world sensor data into large-scale, physics- and photorealistic urban simulation environments, enabling closed-loop testing and policy training for urban autonomous navigation systems. • A powerful 4D long-tail scenario generation engine capable of synthesizing diverse, realistic, and controllable dynamic scenarios from text descriptions, supporting systematic closed-loop testing and training in simulation. • A mixed-reality capability that seamlessly integrates virtual long-tail scenarios into real-world sensor streams, enabling safe and effective closed-course evaluation. • Demonstrated improvement in autonomous navigation reliability under edge and long-tail scenarios, targeting ≥10% closed-loop metrics improvement, validated through mixed-reality closed-course testing. • **Commercialization Potential:** The proposed Real-to-Sim-to-Real (R2S2R) technology establishes a strong foundation for commercialization by enabling **cost-efficient, low-risk validation and reliability improvement for autonomous mobility and robotics systems**. For Korean SMEs, the technology can be productized and integrated in several commercially viable forms: • Simulation and validation software products, including long-tail scenario libraries, 4D scenario generation engines, and simulation plugins for autonomous driving and robotics. • Autonomous driving and ADAS development toolchains that enable differentiated validation, testing, and safety assessment for rare and safety-critical scenarios. • Mixed-reality testing and safety evaluation services for low-risk closed-course validation and pre-deployment assessment of autonomous vehicles and delivery robots. • Platform technologies for rapid localization and customization, converting region-specific driving data into simulation environments and long-tail scenarios to accelerate deployment. By reducing dependence on extensive real-world testing, the proposed technology significantly lowers development cost, shortens iteration cycles, and accelerates time-to-market, making it particularly attractive for SME-led commercialization. **Broader Impacts:** • Improves safety and reliability of autonomous systems under rare but high-risk situations. • Establishes long-term GT-Korean SME collaboration and technology transfer in AI, robotics, and advanced mobility. • Enhanced competitiveness of Korean SMEs in autonomous mobility, robotics, and simulation software markets.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization